



NTNU – Trondheim
Norwegian University of
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Department of Information Security and Communication Technology

Examination paper for TTM4135 Applied Cryptography and Network Security

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Permitted examination support material: (D) No printed or hand-written support material is allowed. A specific basic calculator is allowed.

Other information: –

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Checked by:

Date

Signature

TTM4135 August exam 2024:
Outline answers

Exercise 1 Multiple choice questions

- | | | | | |
|-----|---|---|---|---|
| 1. | (a) ☒ | (b) ☒ | (c) ☐ | (d) ☐ |
| 2. | (a) ☐ | (b) ☒ | (c) ☐ | (d) ☐ |
| 3. | (a) ☒ | (b) ☐ | (c) ☐ | (d) ☐ |
| 4. | (a) ☐ | (b) ☒ | (c) ☐ | (d) ☐ |
| 5. | (a) ☐ | (b) ☒ | (c) ☒ | (d) ☒ |
| 6. | (a) ☐ | (b) ☐ | (c) ☒ | (d) ☐ |
| 7. | (a) ☒ | (b) ☐ | (c) ☐ | (d) ☐ |
| 8. | (a) ☒ | (b) ☐ | (c) ☐ | (d) ☒ |
| 9. | (a) ☒ | (b) ☐ | (c) ☐ | (d) ☐ |
| 10. | (a) ☐ | (b) ☐ | (c) ☐ | (d) ☒ |
| 11. | (a) ☐ | (b) ☐ | (c) ☐ | (d) ☒ |
| 12. | (a) ☐ | (b) ☒ | (c) ☐ | (d) ☐ |
| 13. | (a) ☐ | (b) ☐ | (c) ☐ | (d) ☒ |
| 14. | (a) ☒ | (b) ☐ | (c) ☐ | (d) ☐ |
| 15. | (a) ☐ | (b) ☐ | (c) ☒ | (d) ☐ |
| 16. | (a) ☐ | (b) ☐ | (c) ☐ | (d) ☒ |
| 17. | (a) ☐ | (b) ☐ | (c) ☒ | (d) ☐ |
| 18. | (a) ☐ | (b) ☒ | (c) ☐ | (d) ☐ |
| 19. | (a) ☐ | (b) ☒ | (c) ☐ | (d) ☐ |
| 20. | (a) ☐ | (b) ☐ | (c) ☒ | (d) ☐ |
| 21. | (a) ☒ | (b) ☐ | (c) ☐ | (d) ☐ |
| 22. | (a) ☐ | (b) ☐ | (c) ☒ | (d) ☐ |
| 23. | (a) ☒ | (b) ☐ | (c) ☐ | (d) ☐ |
| 24. | (a) ☐ | (b) ☒ | (c) ☐ | (d) ☐ |

- | | | | | |
|-----|---|---|---|------------------------------|
| 25. | (a) ☐ | (b) ☐ | (c) ☒ | (d) ☐ |
| 26. | (a) ☐ | (b) ☒ | (c) ☐ | (d) ☐ |
| 27. | (a) ☐ | (b) ☐ | (c) ☒ | (d) ☐ |
| 28. | (a) ☐ | (b) ☒ | (c) ☐ | (d) ☐ |
| 29. | (a) ☒ | (b) ☐ | (c) ☐ | (d) ☐ |
| 30. | (a) ☒ | (b) ☐ | (c) ☐ | (d) ☐ |

Exercise 2 Written answer questions

1. (a) $\phi(\omega) \cdot \omega$ – since we must have $\gcd(a, \omega) = 1$, there are $\phi(\omega)$ possible values for a , and ω possible values for b .
 (b) For $\omega = 28$, $\phi(\omega) \cdot \omega = 336$.
 (c) 19.
2. (a) $P_t = C_t \oplus O_t$.
 (b) No (but keystream can be computed in advance)
 (c) No
 (d) A one-bit change in the ciphertext produces a one-bit change in the plaintext at the same location.
3. (a) i. If $\sigma = \text{Sig}(m, K_s)$, then $\text{Ver}(m, \sigma, K_v) = \text{True}$, for any matching signing/verification keys
 ii. It is computationally infeasible for anyone without K_s to construct m and σ such that $\text{Ver}(m, \sigma, K_v) = \text{True}$.
 (b) Can forge signatures: for any two m, m' , where an attacker can obtain the respective signatures σ, σ' , the signature $\sigma \cdot \sigma'$ is a valid signature for the message $m \cdot m'$.
 (c) Hash the message before inputting it into the signature algorithm.
4. (a) They can list all values, or use Fermat's little theorem – all answers are accepted here.
 (b) $b = 4$. Partial credit if they explain methodology/ how to get the answer/ etc., even if the correct value is not recovered.
5. (a) The client doesn't authenticate themselves, and so the server cannot verify their identity.
 (b) – Advantages: flexibility, backwards compatibility.
 – Disadvantage: downgrade attacks.
 (c) The advantage is that we can have a faster connection. The disadvantage is that it can enable certain types of replay attacks.